introduced mission adaptability, allowing drones to switch between reconnaissance, surveillance, and research roles with minimal modification.

Figure 3.2 illustrates the principles and components of second-generation drones, highlighting the integration of advanced electronic systems, GPS navigation, and real-time communication links that enabled semi-autonomous flight. The figure shows the central role of the flight control unit (FCU), which processes input from the inertial measurement unit (IMU) and GPS module to maintain stability, altitude, and directional control. It also depicts the communication and telemetry system, which provides a two-way data link between the drone and the ground control station (GCS) for live video, telemetry, and command exchange. The airframe and propulsion system, built from lightweight composite materials, ensure endurance and maneuverability, while the payload module houses sensors or cameras for reconnaissance and monitoring missions. Overall, the figure demonstrates how the coordinated operation of navigation, control, propulsion, and communication systems transformed second-generation drones into reliable, multi-role aerial platforms capable of precise and efficient mission execution.